

# ADVERSARIAL ML: ROBUSTNESS

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## 1 INTRODUCTION

Machine learning models are susceptible to adversarial examples, inputs with carefully crafted, often imperceptible perturbations that cause misclassification. In this task, we train an image classifier on 50,000 labeled images ( $3 \times 32 \times 32$ , 9 classes) that achieves high accuracy on both clean and adversarially perturbed inputs. The unified score  $\text{Score} = 0.5 \times \text{clean accuracy} + 0.5 \times \text{robust accuracy}$  reflects the fundamental tradeoff between standard and adversarial performance. The attack used for evaluation is unknown, requiring genuine robustness rather than overfitting to a specific threat model.

## 2 APPROACH

**Architecture.** We use a standard `resnet18` with only the final fully-connected layer replaced to output 9 classes. Initializing from ImageNet-pretrained weights (`IMAGENET1K_V1`) significantly outperformed training from scratch (+0.05 unified score), as pretrained features provide strong representations that adversarial training can refine. We found that modifying `conv1` (e.g., changing the  $7 \times 7$  kernel to  $3 \times 3$ ) causes submission rejection, since the server reconstructs the model from the official architecture.

**Training objective.** We adopt TRADES (2):  $\mathcal{L} = \mathcal{L}_{\text{CE}}(f(x), y) + \beta \cdot \text{KL}(f(x) \| f(x_{\text{adv}}))$ , where the first term ensures clean accuracy and the second encourages prediction consistency under perturbation. We set  $\beta = 4.0$  after experimenting with 3.0 (too much clean bias: 80.9% clean, 20.7% robust) and 6.0 (training instability when combined with AWP). Adversarial examples are generated via 7-step PGD on the KL objective with  $\epsilon = 8/255$  and step size  $\alpha = 2/255$ .

**Training details.** SGD with Nesterov momentum (0.9), weight decay  $5 \times 10^{-4}$ , batch size 256, and AMP. Learning rate warms up linearly to 0.1 over 5 epochs, then follows cosine annealing over 150 epochs. Label smoothing of 0.1 is applied to the CE term. Data augmentation: random cropping (padding 4, reflect) and horizontal flipping. We train on all 50,000 samples to maximize data utilization.

**Checkpointing against robust overfitting.** Best robust accuracy often occurs before training ends due to robust overfitting (3). We save the best model (evaluated via PGD-20 every 5 epochs on a 5,000-sample subset), which consistently improved our submitted score over the final-epoch model.

## 3 RESULTS

### Key findings:

- (1) Pretrained ImageNet weights gave the largest single improvement (+0.05 unified score), providing strong feature representations that adversarial training could refine.
- (2) Adversarial Weight Perturbation (AWP) caused severe training instability, loss spiked from 1.6 to 3.9, and did not outperform pure TRADES.
- (3) Stochastic Weight Averaging (SWA), when started during high-LR phases, produced degenerate models (score 0.10 vs base 0.56); it requires careful scheduling after LR decay.
- (4) Best model checkpointing captured the peak before robust overfitting, improving the submitted score by +0.007 over the final epoch model.

| Ver. | Configuration                                           | Local | Leaderboard  |
|------|---------------------------------------------------------|-------|--------------|
| v1   | ResNet18 (modified conv1), TRADES+AWP, $\beta=6$        | 0.548 | rejected     |
| v2   | ResNet18 (standard), TRADES+AWP, $\beta=6$ , LR=0.05    | 0.547 | 0.550        |
| v3   | ResNet50 (standard), TRADES+SWA, $\beta=6$              | 0.563 | 0.570        |
| v4   | ResNet18 (pretrained), TRADES, $\beta=4$ , 120ep        | —     | 0.607        |
| v5   | ResNet18 (pretrained), TRADES, $\beta=4$ , 200ep + ckpt | 0.619 | <b>0.614</b> |

Table 1: Leaderboard progression. Pretrained initialization and best-model checkpointing provided the largest gains. “ckpt” = best-model checkpointing via PGD-20.

## 4 CONCLUSION

Our best model achieves a unified score of 0.614 on the public leaderboard using TRADES adversarial training with a pretrained ResNet-18 and best model checkpointing. The practical implications of adversarial vulnerability extend well beyond academic benchmarks. In autonomous driving, adversarially perturbed road signs could trigger dangerous maneuvers. In medical imaging, imperceptible modifications to diagnostic scans could alter clinical decisions. In biometric security, adversarial perturbations could bypass authentication systems. These threats are compounded by the transferability of adversarial examples, attacks crafted against one model often fool others, including black-box systems whose architectures are unknown. Our experiments highlight two key challenges for deploying robust models in practice: first, improving adversarial robustness consistently reduces clean accuracy, requiring practitioners to navigate this tradeoff carefully; second, robust overfitting causes model quality to degrade during training, making checkpoint selection critical. Addressing these challenges remains essential for building trustworthy ML systems in safety critical applications.

**Code:** [https://github.com/MARKST-47/TML-26/tree/main/tml26\\_task3](https://github.com/MARKST-47/TML-26/tree/main/tml26_task3)

## REFERENCES

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